

Triple-AI/ML and the Six-Nation Bound

An Axiomatic Theory of Epistemic Compression

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops an axiomatic model of geopolitical compression generated by advanced artificial-intelligence systems. The central object is the Triple-AI/ML coalition, a collection of three geographically distinct nations sharing a common epistemic infrastructure. We postulate that the existence of one Triple induces a second Triple, generating a dual informational structure consisting of knowledge and propaganda. Nations outside the dual structure become hinge actors whose informational behaviour is representable by Boolean operations. Under these assumptions, the number of epistemically sovereign nations is bounded above by six.

The paper ends with “The End”

1 Introduction

Artificial intelligence increasingly influences economic allocation, political communication, military planning, and strategic decision making. This paper investigates a limiting theoretical case in which informational autonomy becomes the primary source of power. The objective is not empirical description but axiomatic analysis.

2 Definitions

Definition 2.1. A *nation* is a territorially sovereign political actor.

Definition 2.2. An *epistemically sovereign nation* is a nation capable of generating independent informational states.

Definition 2.3. A *Triple* is a set

$$T = n_1, n_2, n_3,$$

where the three elements are geographically distinct nations.

Definition 2.4. A *hinge nation* is a nation belonging to neither of the two Triples.

3 Axioms

Axiom 3.1 (Existence). At least one Triple exists.

Axiom 3.2 (Epistemic Induction). The existence of a Triple implies the emergence of a second Triple.

Axiom 3.3 (Informational Duality). The two Triples generate informational outputs classified as knowledge or propaganda.

Axiom 3.4 (Boolean Hinge Principle). Every hinge nation receives information exclusively through outputs generated by the two Triples.

Axiom 3.5 (Ontological Reduction). Any actor whose informational behaviour admits a complete Boolean representation lacks epistemic sovereignty.

4 Informational Structure

Let

$$\mathcal{I} : T_1 \cup T_2 \rightarrow K, P$$

denote the informational classification operator.

Here (K) denotes knowledge and (P) denotes propaganda.

For a hinge nation (h), define the reduction operator

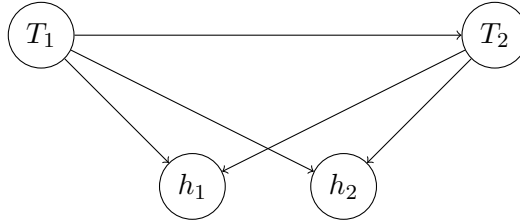
$$\mathcal{R}(h) = b_h,$$

where

$$b_h \in 0, 1.$$

Thus every hinge nation is represented by a Boolean informational state.

5 Epistemic Geometry



The figure illustrates informational dependence. The hinge nations consume information generated by the two Triples but do not create independent epistemic structures.

6 Main Results

Lemma 6.1. *Each Triple contains exactly three nations.*

Proof. This follows directly from the definition of a Triple. □

Lemma 6.2. *The union of two Triples contains at most six nations.*

Proof. Each Triple contains three nations. Therefore

$$|T_1| + |T_2| = 6.$$

The maximum cardinality occurs when the Triples are disjoint. □

Theorem 6.3 (Six-Nation Bound). *The number of epistemically sovereign nations satisfies*

$$N_E \leq 6.$$

Proof. By the previous lemma,

$$|T_1 \cup T_2| \leq 6.$$

Every nation outside $(T_1 \cup T_2)$ is a hinge nation.

By the Boolean Hinge Principle and Ontological Reduction, hinge nations are not epistemically sovereign.

Therefore every epistemically sovereign nation belongs to $(T_1 \cup T_2)$.

Hence

$$N_E \leq |T_1 \cup T_2| \leq 6.$$

□

Corollary 6.4. *The number of territorial states may exceed six even when the number of epistemically sovereign states does not.*

7 Economic Interpretation

Let (g_i) denote the economic output of nation (i) .

The output generated by epistemically sovereign nations is

$$G_E = \sum_{i \in T_1 \cup T_2} g_i.$$

The model therefore predicts concentration of informational capital.

8 Financial Interpretation

Let (r_t) denote a financial return.

Suppose market expectations satisfy

$$E[r_t] = \alpha E_K[r_t] + (1 - \alpha) E_P[r_t],$$

where

$$0 \leq \alpha \leq 1.$$

Knowledge and propaganda become competing expectation operators.

9 Political Science and International Relations

Classical theories emphasize military, demographic, and territorial resources.

The present framework emphasizes informational autonomy.

Define strategic capability by

$$S = M^\beta I^{1-\beta},$$

where (M) denotes military capability and (I) denotes informational capability.

As AI systems become increasingly important,

$$\beta \rightarrow 0.$$

Informational capability becomes dominant.

10 Warfare Economics

Let

$$W = f(C, D, A),$$

where

- (C) is computational capacity,
- (D) is data,
- (A) is algorithmic sophistication.

The model predicts a gradual transition from industrial warfare toward informational warfare.

11 Machine Learning Interpretation

The dual-Triple structure resembles an adversarial learning architecture.

One Triple produces informational hypotheses.

The second produces competing informational hypotheses.

Hinge nations perform classification and selection.

The resulting system resembles a repeated adversarial learning process.

12 Statistical Interpretation

Define

$$Y = \begin{cases} 1, & \text{knowledge,} \\ 0, & \text{propaganda.} \end{cases}$$

The informational environment becomes a binary classification problem.

The reduction concerns epistemic dimensionality rather than geography.

13 Discussion

The theory distinguishes territorial multiplicity from epistemic multiplicity.

Many territorial states may exist simultaneously.

However, under the stated axioms, informational autonomy is concentrated within a dual-Triple structure.

The Six-Nation Bound therefore applies to epistemically sovereign actors rather than to territorial states themselves.

14 Conclusion

Starting from five axioms, we derived an upper bound of six epistemically sovereign nations.

The mechanism is epistemic compression: actors external to the dual-Triple structure become informationally reducible and therefore lose epistemic autonomy.

Whether such structures emerge in reality remains an empirical question.

Field	Principal Consequence
Economics	Concentration of informational capital
Finance	Competing expectation operators
Political Science	Epistemic sovereignty
International Relations	Balance of information
Warfare	Informational conflict
AI/ML	Adversarial learning structure
Statistics	Binary classification

Table 1: Consequences of the Triple-AI/ML framework.

References

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Glossary

Triple

A three-nation epistemic coalition.

Knowledge

Informational output classified as epistemically productive.

Propaganda

Informational output classified as strategically manipulative.

Hinge Nation

A nation external to both Triples.

Boolean Hinge Principle

The assumption that hinge behaviour is representable by Boolean operations.

Ontological Reduction

Loss of epistemic sovereignty through complete Boolean representation.

Epistemic Compression

Reduction of many territorial actors into a small number of informationally autonomous actors.

Six-Nation Bound

The theorem stating that the number of epistemically sovereign nations does not exceed six.

The End